

Ahmed Ikram | AI Engineering

Dundee, Scotland | +44 7421 996048 | ahmedikram30@gmail.com | [LinkedIn](#) | [GitHub](#) | [Online Portfolio](#)

PROFILE

Final-year BSc Computer Science (Data Science & AI) student at the University of Dundee, on track for a First-Class degree, graduating June 2027. Builds production LLM systems end-to-end - agentic RAG with multi-signal confidence fusion, LangGraph tool-calling agents, and the LLMops platforms around them: data pipelines, execution-based evaluation, statistical promotion gating, and OpenAI-compatible serving. Recent work: a RAG diagnostic assistant backing a **99.8%**-accurate anomaly detection engine, and an LLMops platform that lifted a fine-tuned Qwen3-14B's SWE-bench fix rate **7.0×** over the base model under a statistically-gated promotion pipeline. Seeking a graduate AI Engineering role - or an adjacent Applied ML / MLOps / LLMops position - building and evaluating reliable LLM systems in production.

EDUCATION

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- On track for a **First-Class Honours** degree. Expected Graduation: **June 2027**

High School of Dundee

Aug 2017 - June 2023

- Advanced Higher Computer Science (C), Higher Maths and English (A), Higher Chemistry (B) and Higher Physics (C)

TECHNICAL SKILLS

LLM & Generative AI: PyTorch · Hugging Face Transformers · QLoRA · LoRA · PEFT · TRL · Unsloth · vLLM · AWS Bedrock · Prompt & Context Engineering

AI Systems, Agents & Evaluation: LangGraph · LangChain · Agentic RAG · ChromaDB · Ollama · Execution-Based & LLM-as-Judge Evaluation · SWE-bench · Cross-Encoder Reranking · LangSmith · Langfuse · Multimodal (Vision-Language) Extraction · NLP

MLOps & Model Serving: MLflow · Weights & Biases · Airflow · Evidently AI · FastAPI · Modal · SageMaker

Languages: Python · SQL · TypeScript · JavaScript · Bash · Rust

Data & Backend: PostgreSQL · Redis · Kafka · MySQL · MongoDB · Pandas · NumPy · Pydantic

Cloud, Infrastructure & Testing: AWS (ECS Fargate, EC2, S3, RDS, ElastiCache, IAM, SageMaker, Bedrock, CloudFront, CloudWatch, Secrets Manager) · Google Cloud · Terraform · Docker · GitHub Actions CI/CD · OIDC · Pytest · mypy · Ruff

PROJECTS

ATM Log Aggregation, Analysis & Diagnostics Platform

FastAPI · XGBoost, Isolation Forest · MLflow, SageMaker · LangChain · ChromaDB · Kafka · Redis · Docker · AWS

- Backend and ML engineering lead on a 7-person Agile team delivering a diagnostics platform for NCR Atleos - a real industry stakeholder, not a solo portfolio project; independently built the 3-layer detection engine, Agentic RAG diagnostic assistant, full MLOps/AWS infrastructure and a 1,402-test suite (923 pytest, 394 vitest, 10 playwright E2E, 75 terraform assertions) - all post-deadline on top of the team delivered foundation
- Agentic RAG** diagnostic assistant over ChromaDB + Ollama: **context engineering** via metadata-filtered retrieval and cross-encoder reranking lifts retrieval relevance +5–15% retrieval relevance over cosine similarity alone; a 4-stage reasoning loop (self-consistency across 3 parallel samples verbalized confidence, reflexion self-critique, citation grounding) drives a Platt-calibrated 4-signal confidence fusion (retrieval 30% + consistency 25% + verbalized 25% + grounding 20%); parallelising self-consistency cut end-to-end latency to 11-23s uncached and under 100ms on a 300s Redis cache hit, backed by a 3-provider LLM fallback chain and a prompt-injection filter
- 3-layer anomaly detection engine (**XGBoost + Isolation Forest** ensemble, rolling 20-window Z-score, 7 deterministic heuristics) runs every 30s against a live Kafka-fed event stream on 49 engineered features across 7 groups, achieving $99.8\% \pm 0.1\%$ StratifiedKFold CV accuracy across 8 classes and 97.3% Isolation Forrest precision via a 200-point Youden's J threshold sweep ($F1=0.7008$ at -0.51009); predictions cross-checked against a deployed AWS SageMaker endpoint with all runs and 7 artifacts versioned in MLflow under champion aliases
- Full **AWS** infrastructure (ECS Fargate, RDS, S3, SageMaker, Secrets Manager) provisioned via Terraform with 3 OIDC authenticated GitHub Actions pipelines and zero static credentials

StockLens FinTech Spending & Investment App

LangGraph · AWS · PyTorch · Optuna · FastAPI · Airflow · MLflow · Evidently AI · Tesseract · PostgreSQL · Redis · Terraform

- 2-node **LangGraph ReAct** agent (explicit StateGraph, not the create_react_agent wrapper) exposing 16 tools across 7 categories to AWS Bedrock with SSE-streamed responses; two-tier Redis (7-day TTL) & PostgreSQL persistence survives server restarts with full conversation-history reconstruction; correctness judged by an LLM-as-judge in CI with LangSmith tracing at a 10% sample rate
- Confidence-gated 4-stage **multimodal NLP** extraction cascade for receipt understanding: Tesseract OCR + heuristic field scoring escalates through a Bedrock Vision-LLM (image → structured JSON) to a text-only LLM fallback and a degraded final tier whenever confidence dips below threshold (overall <0.7, OCR <0.6, unverified merchant, or a reconciliation mismatch); RapidFuzz merchant matching (≥80% → 0.95 confidence) and a 24h Redis cache keyed on receipt hash avoid re-invoking the vision model on repeat scans
- A global LSTM forecaster - exposed to the agent as a callable tool - predicts 5-day directional price movement across the S&P 500, reaching **51.6%** accuracy against a **33%** random baseline (0.97 simulated Sharpe)
- Weekly Airflow-orchestrated retraining pipeline (6-year OHLCV inget, MLflow-tracked training, champion/challenger promotion gated at >2pp directional-accuracy improvement) with Evidently AI drift monitoring (PSI/KS/JSD) alerting to CloudWatch
- Validated by 1,415 backend and 822 frontend tests at a 90% coverage gate, enforced by a 9-parallel-job CI pipeline and a 7-stage OIDC-authenticated CD pipeline on Terraform-provisioned AWS infrastructure (14 modules, 193 resources)

SWE-Qwen Automated LLM Fine Tuning, Evaluation & Serving Platform

Python · PyTorch · QLoRA · Unsloth · Hugging Face · Modal · vLLM · Weights & Biases · Langfuse · FastAPI · Terraform · Google Cloud · GitHub Actions

- Model-agnostic **LLMOps platform** spanning schema-validated data preparation, QLoRA fine-tuning, execution-based evaluation, statistical champion/challenger promotion, and OpenAI-compatible inference serving - config-driven model swapping and versioned GCS/W&B artifacts shared across every stage
- Six-stage data pipeline (Pydantic schema validation, 6 automated quality gates, exact & semantic dedup, per-repo stratified 80/10/10 split) turned 20,477 raw SWE-bench issues into a 17,456-example high-signal corpus across 46 repos with zero cross-repo leakage - yielding 14,833 tokenized training examples and a 2,313-instance golden evaluation set carved out before tokenization
- Execution-based evaluation harness materialises each repo inside its official SWE-bench Docker image and runs real FAIL_TO_PASS/PASS_TO_PASS test suites - not an LLM-graded proxy; scored 3 QLoRA Qwen3-14B variants (4-bit NF4, Unsloth + FlashAttention on Modal A100-80GB) with the champion reaching 17.20% F2P (95% Wilson CI 11.1-25.8%, 7.0× over base, McNemar p<1e-6) at 90.10% P2P
- Promotion gated behind a 4-condition statistical test (absolute F2P/P2P floors, CI-lower-bound > 0, ≤2pt P2P-regression ceiling) so a challenger can never self-certify a win; deployed as an OpenAI-compatible API on Modal scale-to-zero infrastructure with per-request LoRA adapter swapping via vLLM + AWQ - zero engine restarts, \$0 idle cost
- Provisioned via Terraform with GCP Workload Identity Federation (zero static credentials); 4 GitHub Actions workflows - including a per-PR execution-based smoke gate where only main-branch runs can write the regression baseline - and 1,456 tests with strict mypy typing; every run traced via W&B and Langfuse (10% sampled)

CERTIFICATES

AWS Academy Graduate - Machine Learning Foundations

Feb 2025

AWS Academy Graduate - Cloud Foundations

Mar 2024

Foundations of Azure AI: Concepts, Capabilities, and Implementation

Oct 2024

EXPERIENCE

Beaully Mini Market, Dundee, Scotland

Retail Assistant

September 2023 - Present

- Diagnosed and resolved Point-of-Sale and stock system failures rapidly to minimise service disruption

Zulus Peri Peri Grill, Dundee, Scotland

Front of House Worker

June 2021 - September 2023

- Trained 5 new staff members; managed competing priorities across dine-in, takeaway, and delivery orders under sustained pressure